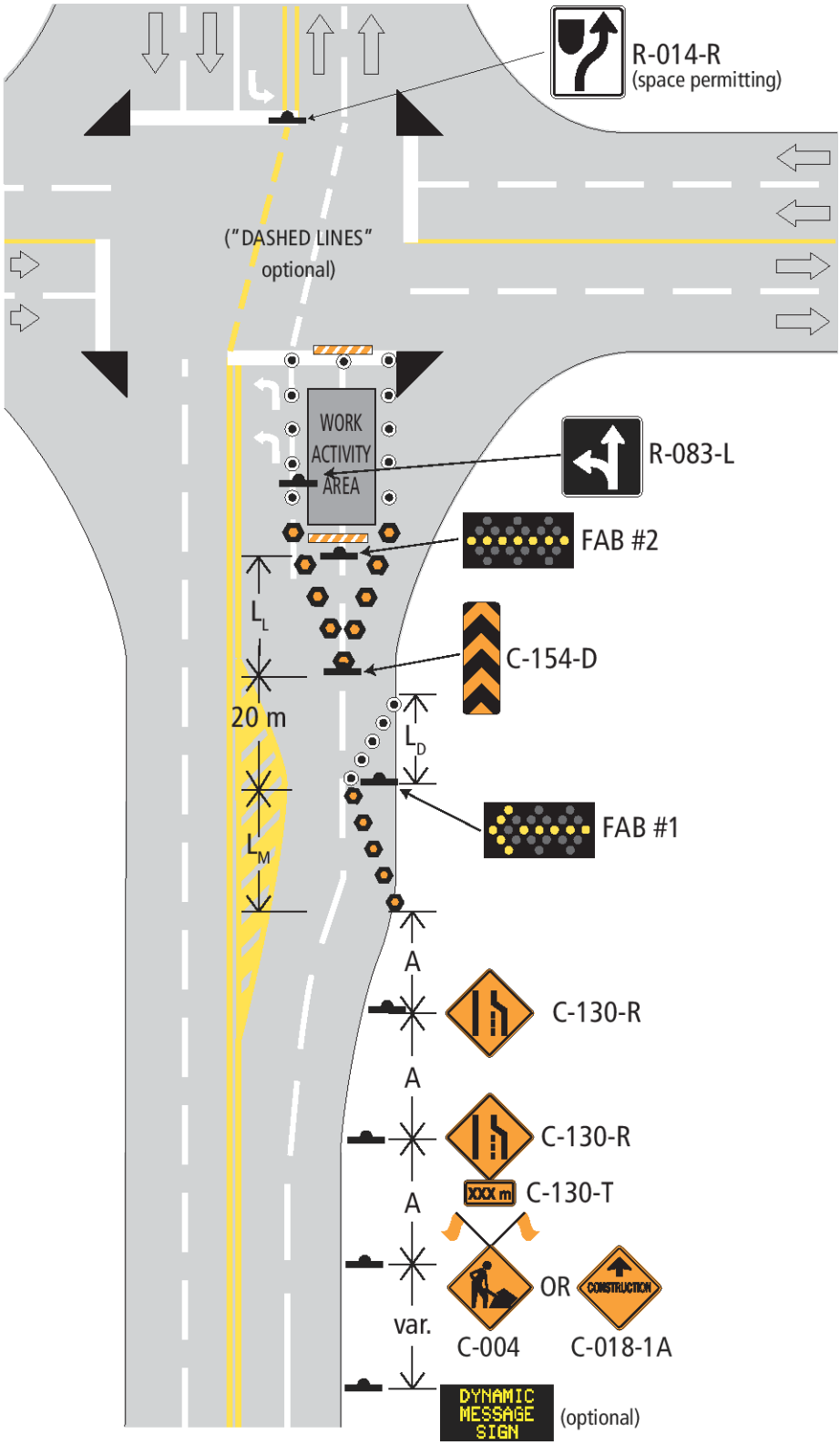
Figure 11.11: Two Lanes Closed (Near Side) – Multilane Intersection – Short and Long Duration



11.12 Two-Lane Closure – Multilane Intersection – Short and Long Duration

Purpose:

This layout shows the typical setup for a double lane closure on a multilane roadway where the closure occurs both upstream (near side) and downstream (far side) of a signalized or stop-controlled intersection, but not through the intersection itself.

Traffic from the closed lanes is re-routed across the centreline into the opposing travel lane through a median crossover, and the number of lanes available for opposing traffic is reduced using lane drops. The placement and maintenance of channelizing devices is critical.

Standard:

- Flashing arrow boards (FABs #1, #2, #3, and #4) in arrow mode are required in advance of the work area on the major road, and shall be positioned inside the approaching taper for each closed lane or lane shift.
- Two-Way Traffic Ahead C-132 signs shall be used in advance of locations where two-way traffic begins.
- Two-Way Traffic R-010 signs shall be used along tangents where two-way traffic has been established.
- A Keep Right R-014-R sign shall be placed on the far side of the intersection to direct drivers to keep right.

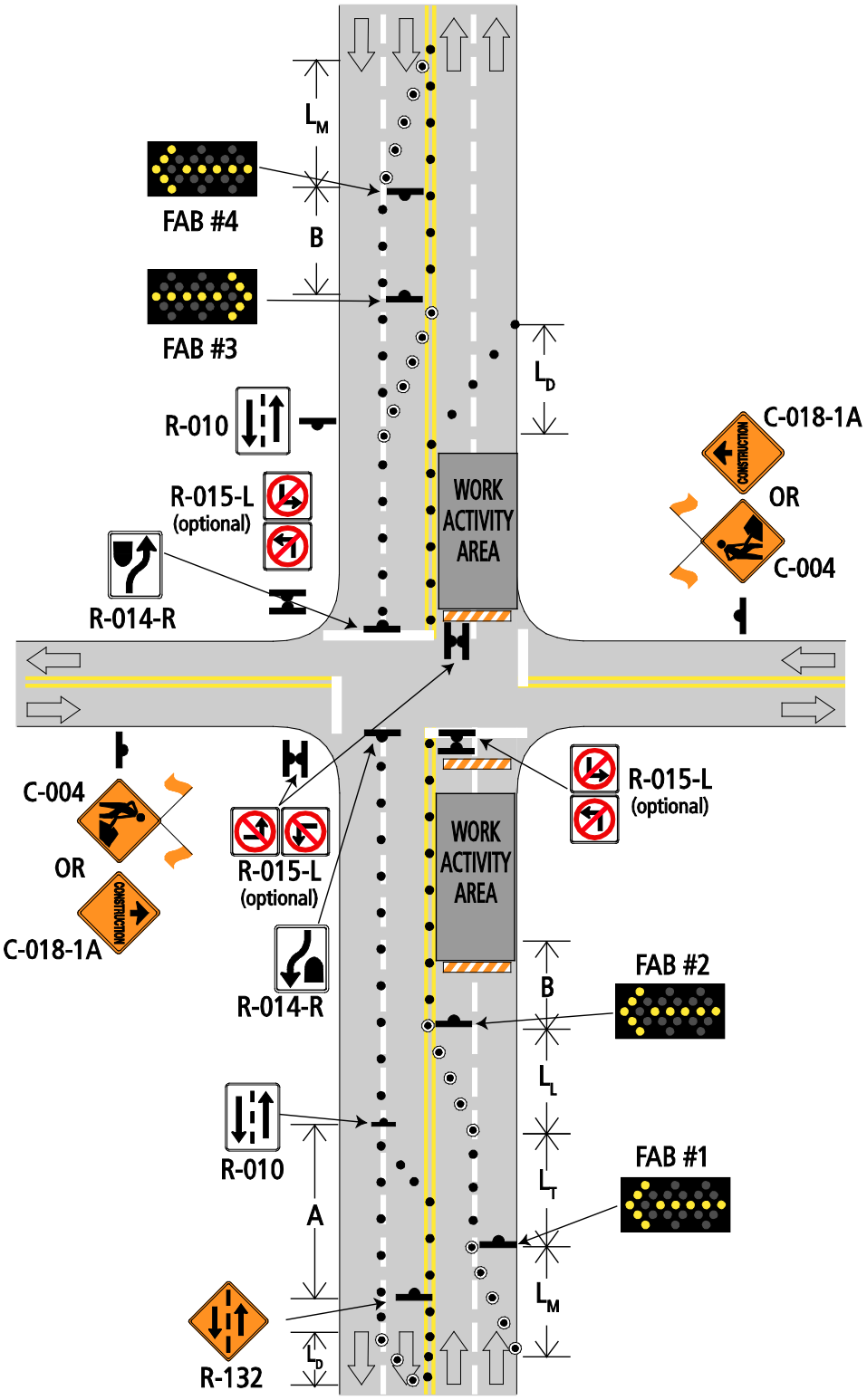
Guidance:

- Median-mounted signs matching the shoulder-mounted signs should be used where space allows.

Options:

- Temporary dashed pavement marking may be added across the intersection to guide traffic through the intersection into the downstream lane.
- If the speed limit is ≤ 60 km/h:
  - A Lane Closure Arrow C-053 sign may replace FAB #2 and FAB #3

Figure 11.12: Two-Lane Closure – Multilane Intersection – Short and Long Duration



11.13 Multiple Lane Closure (Far Side) with Dedicated Left-Turn Lane – Multilane Intersection – Short and Long Duration

Purpose:

This layout shows the typical setup for a double lane closure on a multilane roadway when the closure occurs downstream (far side) of an intersection.

Traffic from the closed lanes is merged into the left-turn lane on the upstream (near) side and travels through the intersection into the opposing left-turn lane.

The number of lanes available for opposing traffic is reduced, so the placement and maintenance of channelizing devices is critical.

Standard:

- A Right Lane Closed Ahead C-130-R sign with a C-130-T distance tab is required in advance of a second Right Lane Closed Ahead C-130-R sign.
- A double-sided taper shall be indicated with drums, and a Double Hazard C-154-D marker shall be positioned in advance of the work to direct traffic to the left and right lanes.
- Flashing arrow boards (FABs) shall be positioned as follows:
  - FAB #1 in arrow mode: in the right lane drop taper in advance of the work area
  - FAB #2 in caution mode: inside the work area taper in advance of the work
- A Lane Closure Arrow C-053 sign shall be used in advance of the work area.

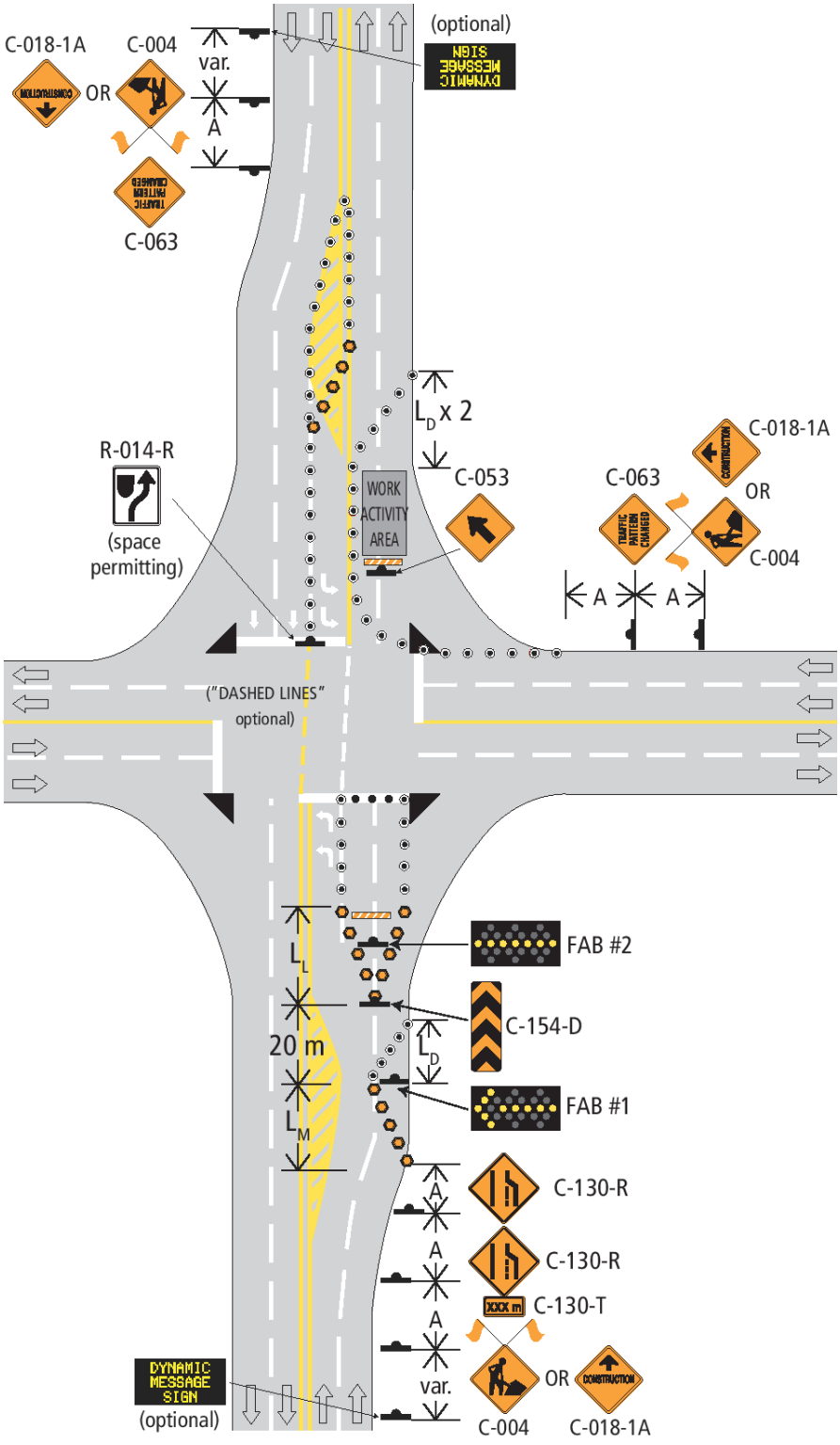
Guidance:

- Traffic moving through the intersection is intended to use the left-turn lane. Signal timings may have to be adjusted in all directions at signalized intersections.
- Median-mounted signs matching the shoulder-mounted signs should be used where space allows.

Options:

- A dynamic message sign may be used in advance of the work.
- Temporary dashed pavement marking may be added across the intersection to guide traffic through the intersection into the downstream lane.
- If the speed limit is ≤ 60 km/h:
  - The upstream Right Lane Closed Ahead C-130-R sign may be omitted and the Crew Working Ahead C-004 sign or Construction Ahead C-018-1A sign moved downstream by Table B Distance A.
  - Tubular markers may be used for leading tapers instead of drums.
  - A Lane Closure Arrow C-053 sign may replace FAB #1.

Figure 11.13: Multiple Lane Closure (Far Side) with Dedicated Left-Turn Lane – Multilane Intersection – Short and Long Duration



11.14 Midblock Sidewalk Detour – Multilane Roadway – Short and Long Duration

Purpose:

This layout shows the typical setup for a sidewalk detour that uses the outside lane of a multilane roadway or the parking lane of a two-lane, two-way roadway.

Standard:

- The pedestrian detour shall be clearly delineated so that the route is apparent to both pedestrians and motorists.
- A Right Lane Closed Ahead C-130-R sign with a C-130-T distance tab is required in advance of a second Right Lane Closed Ahead C-130-R sign.
- The lane closure taper shall be delineated with drums.
- Flashing arrow boards (FABs) in arrow mode shall be used inside the approaching taper in advance of the work area and before the pedestrian route, and for speeds ≥ 70 km/h.

Guidance:

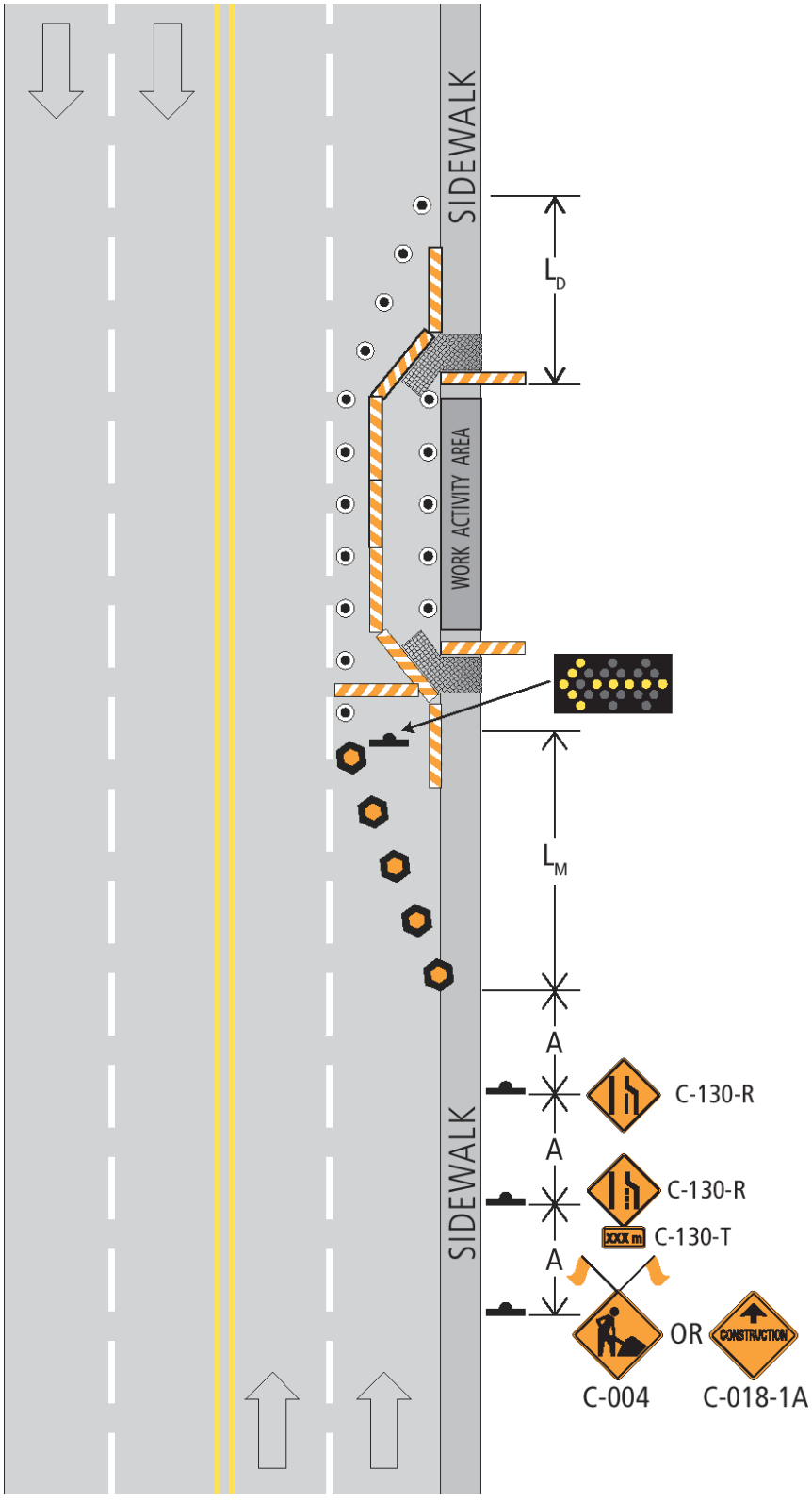
- Clear delineation of the pedestrian route can be achieved in various ways, such as using closely-spaced tubular markers, barricades, fencing, or temporary barriers.
- Temporary pedestrian access routes should be 1.5 metres wide or wider, with a minimum width of 1.2 metres where constraints exist.
- A hard, temporary walking surface should cover rough, soft, or uneven ground.
- At abrupt elevation changes (e.g., between sidewalk and road levels), a temporary curb ramp should be provided, with anti-slip treatment and a slope of 12:1 (8%) or less.
- When crosswalks, sidewalks, or other pedestrian facilities are blocked, closed, or relocated, temporary facilities should include accessibility features that are consistent with those in the existing pedestrian facility.

Options:

- Instead of a ramp, a boardwalk with a railing may be installed at sidewalk level around the closure for high-volume pedestrian areas or projects that last for several days or more.
- Depending on the type of work and the condition of the site, barricades may have to be supplemented with fencing or other devices to physically prevent pedestrians from straying into the work area (see *Section 4.10.1: Work Zone Fencing*).
- The flashing arrow board (FAB) may be replaced as shown below.

Speed Limit	Workers or Work Vehicles Present	No Workers or Vehicles Present
≤ 60 km/h	Lane Closure Arrow C-053 sign plus 360° flashing light and 4-way flashers.	Lane Closure Arrow C-053 sign plus barricade and Type A yellow warning light.
≥ 70 km/h	No substitution.	

Figure 11.14: Midblock Sidewalk Detour – Multilane Roadway – Short and Long Duration



11.15 Midblock Sidewalk Closure – Short and Long Duration

Purpose:

This layout shows the typical setup for a sidewalk detour that is not able to use the outside lane of a multilane roadway or the parking lane of a two-lane, two-way roadway.

This layout is used where a sidewalk is closed and pedestrians are detoured to another pedestrian facility. It affects only the path that pedestrians must follow, and has no impact on traffic.

Standard:

- The sidewalk shall be closed using a barricade and a Sidewalk Closed C-202 sign on both sides of the work.
- A second barricade with a Sidewalk Closed – Cross Here C-203-L/R sign shall be placed in advance of the closure to advise pedestrians of a suitable alternative crossing.
- The pedestrian detour shall be clearly delineated so that the route is apparent to pedestrians.

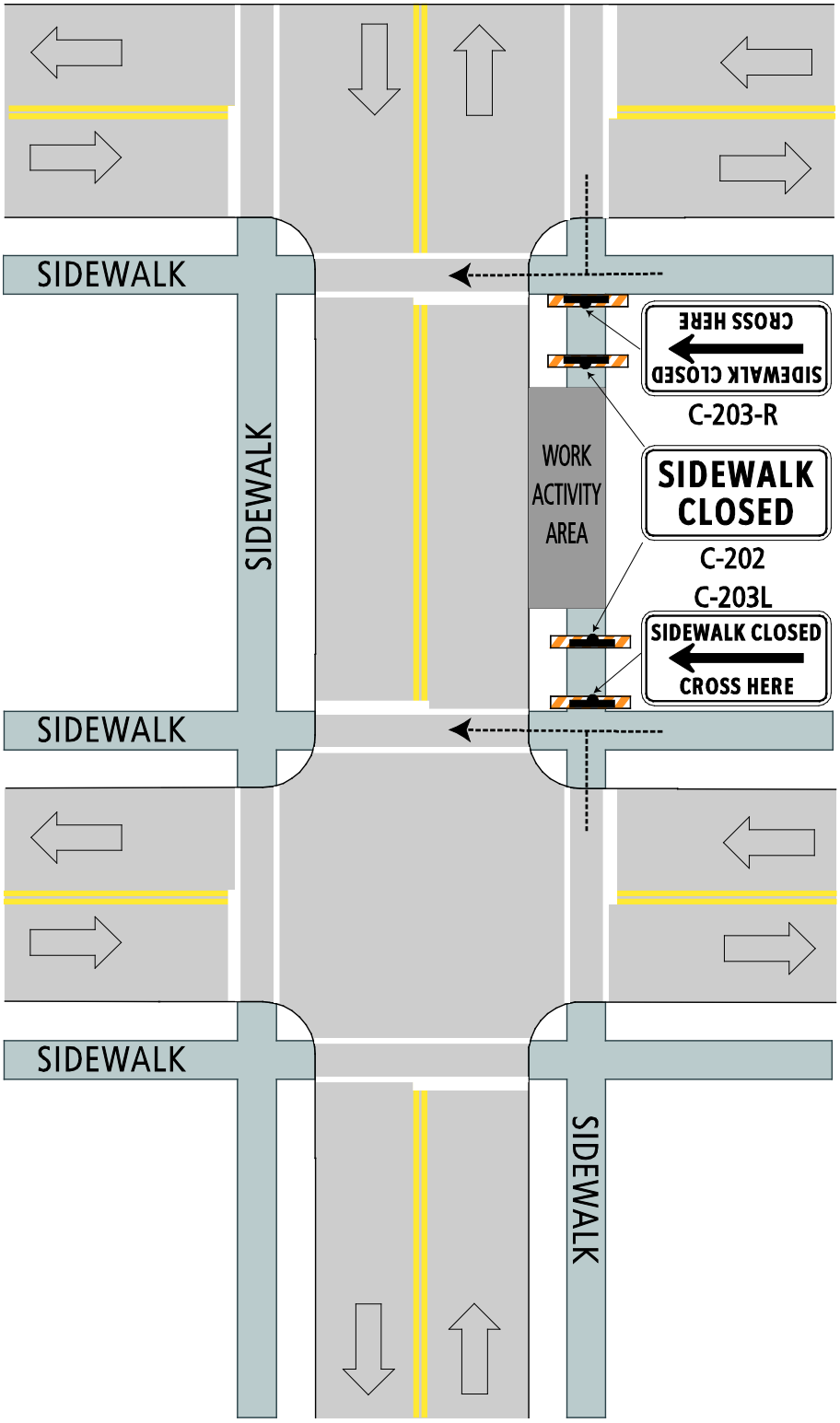
Guidance:

- Pedestrians should be notified of the closure both in advance of the closure and at the closure itself.
- Pedestrians should be advised of the location of an available crosswalk that they can use to cross the road and access a sidewalk on the other side.
- When crosswalks, sidewalks, or other pedestrian facilities are blocked, closed, or relocated, temporary facilities should include accessibility features that are consistent with those in the existing pedestrian facility.
- The roadway may require delineation if the sidewalk work encroaches onto it.

Options:

- Depending on the type of work and the condition of the site, the barricades may have to be supplemented with fencing or other devices to physically prevent pedestrians from straying into the work area (see *Section 4.10.1: Work Zone Fencing*).

Figure 11.15: Midblock Sidewalk Closure – Short and Long Duration



11.16 Sidewalk Detour – Multilane Intersection – Short and Long Duration

Purpose:

This layout shows the typical setup at an intersection and on a corner for a sidewalk detour onto a multilane roadway or a two-lane, two-way roadway that has a parking lane. Where possible, the crosswalks should be kept open.

Standard:

- The pedestrian detour shall be clearly delineated so that it is apparent to both pedestrians and motorists.
- A right lane closure is required for any area where the sidewalk detours into the travel lane:
  - A Right Lane Closed Ahead C-130-R sign and a second C-130-R sign with a C-130-T distance tab are required on the leg of the intersection where traffic is being merged to one lane.
  - In the cross direction, a Right Lane Closed C-030-4A sign is required in advance of the intersection, followed by a Right Lane Must Turn Right R-082-R2 sign to advise that traffic in the outside lane cannot go through the intersection and must turn right.
- The lane closure taper shall be delineated with drums.
- Flashing arrow boards (FABs) in arrow mode shall be used inside the approaching taper in advance of the work area and before the pedestrian route, and for speeds ≥ 70 km/h.

Guidance:

- Pedestrian signals controlling closed crosswalks should be covered or deactivated.
- Clear delineation of the pedestrian route can be achieved in various ways, such as using closely-spaced tubular markers, barricades, fencing, or temporary barriers.
- Temporary pedestrian access routes should be 1.5 metres wide or wider, with a minimum width of 1.2 metres where constraints exist.
- A hard, temporary walking surface should cover rough, soft, or uneven ground.
- At abrupt elevation changes (e.g., between sidewalk and road levels), a temporary curb ramp should be provided, with anti-slip treatment and a slope of 12:1 (8%) or less.
- When crosswalks, sidewalks, or other pedestrian facilities are blocked, closed, or relocated, temporary facilities should include accessibility features that are consistent with those in the existing pedestrian facility.

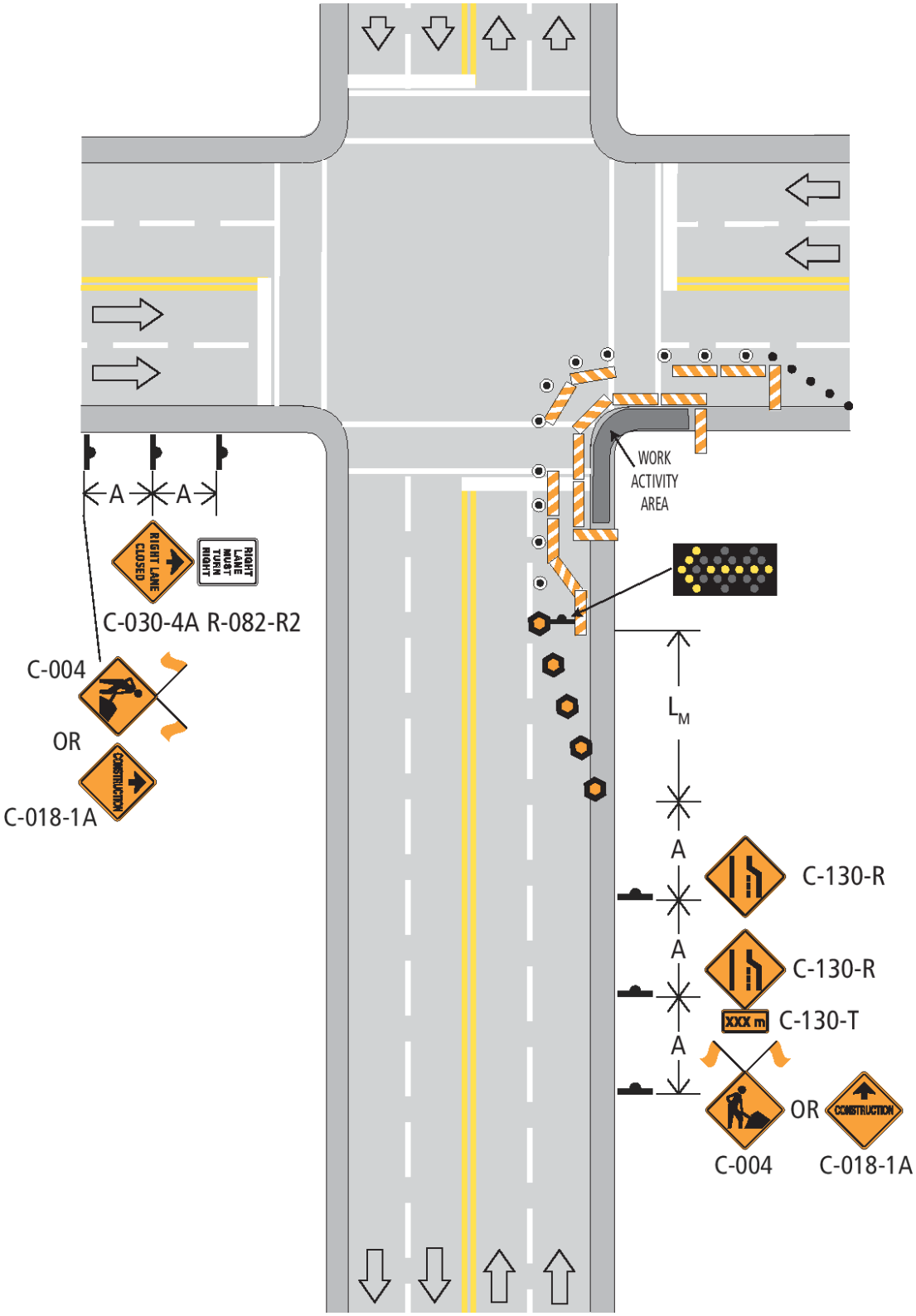
Options:

- A full lane closure may be considered for the right lane of the cross street if high volumes of traffic are attempting to proceed straight through the intersection or if there is confusion about the Right Turn Only signage.
- Instead of a ramp, a boardwalk with a railing may be installed at sidewalk level around the closure for high-volume pedestrian areas or projects that last for several days or more.
- Depending on the type of work and the condition of the site, barricades may have to be supplemented with fencing or other devices to physically prevent pedestrians from straying into the work area (see Section 4.10.1: Work Zone Fencing).
- If the speed limit is ≤ 60 km/h:
  - The upstream Right Lane Closed Ahead C-130-R sign may be omitted and the Crew Working Ahead C-004 sign or Construction Ahead C-018-1A sign moved downstream by Table B Distance A.
  - Tubular markers may be used for leading tapers instead of drums.
- The flashing arrow board (FAB) may be replaced as shown below.

Speed Limit	Workers or Work Vehicles Present	No Workers or Vehicles Present
≤ 60 km/h	Lane Closure Arrow C-053 sign plus 360° flashing light and 4-way flashers.	Lane Closure Arrow C-053 sign plus barricade and Type A yellow warning light.
≥ 70 km/h	No substitution.	

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Figure 11.16: Sidewalk Detour – Multilane Intersection – Short and Long Duration



11.17 Intersection Sidewalk and Crosswalk Closure – Short and Long Duration

Purpose:

This layout shows the typical setup where crosswalks at an intersection are affected by sidewalk or roadside work on one of the intersection corners.

In this situation, the affected crosswalk cannot be used, and a detour in advance of the closure is required to direct pedestrians to another pedestrian facility.

Standard:

- The pedestrian detour shall be clearly delineated so that the route is apparent to both pedestrians and motorists.
- Barricades shall be used to close the both the sidewalk approach and the crosswalk approach to prevent access to the work area.
- Barricades immediately adjacent to the work shall display Sidewalk Closed C-202 signs.
- Additional barricades with Sidewalk Closed – Cross Here C-203-L/R signs shall be placed in advance of the closure to advise pedestrians of a suitable alternative crossing.

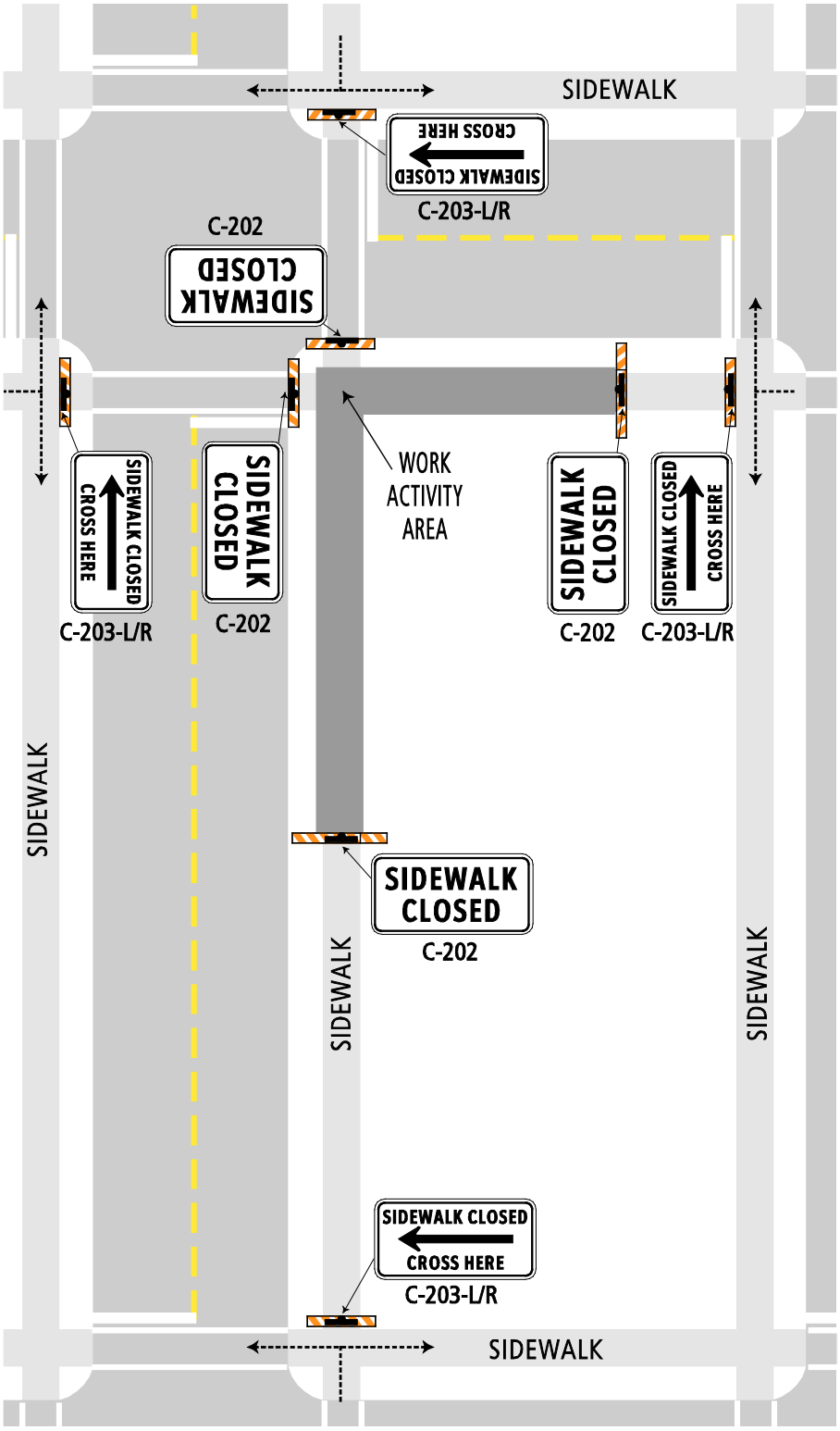
Guidance:

- Pedestrian signals controlling closed crosswalks should be covered or deactivated.
- Pedestrians should be notified of the closure both in advance of the closure and at the closure itself.
- Pedestrians should be advised of the location of an available crosswalk that they can use to cross the road and access a sidewalk on the other side.
- When crosswalks, sidewalks, or other pedestrian facilities are blocked, closed, or relocated, temporary facilities should include accessibility features that are consistent with those in the existing pedestrian facility.
- The roadway may require delineation if the sidewalk work encroaches onto it.

Options:

- A temporary crosswalk may be implemented if there is no nearby pedestrian crossing opportunity.
- If a temporary midblock crosswalk is implemented, curb parking should be prohibited within 30 metres in advance of the midblock crosswalk.

Figure 11.17: Intersection Sidewalk and Crosswalk Closure – Short and Long Duration



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PART B – TRAFFIC CONTROL

Section 12: Traffic Control Layouts – Roundabouts[1]

Contents

Legend, Table A, and Table B 12-1

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12.5 Outer Lane Closure – Multilane Roundabout 12-10

1 See *Section 11: Traffic Control Layouts* – Intersections for information on traffic control layouts for other intersections.

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PART B – TRAFFIC CONTROL

Section 12: Traffic Control Layouts – Roundabouts

LEGEND			
	Flexible Drum		Paint Truck
	Tubular Marker		Escort Truck
	Cone		Chaser Vehicle
	Sign		Vehicle Mounted Rear Crash Attenuator
	Traffic Control Person		
	Work Activity Area		360° Flashing Light
	Work Truck		Portable Traffic Signal
	Shadow Vehicle		Barricade and Fencing
	Shadow Vehicle #1		Flashing Arrow Board (FAB)
	Shadow Vehicle #2		Flashing Arrow Board (FAB) in caution mode
	Buffer Vehicle		Dynamic Message Sign (DMS)

TABLE A – TAPER LENGTHS

Taper Types (m)		Regular Posted Speed Limit (km/h)							
		≤50	60	70	80	90	100	110	120
Merge Taper Length	L_M	35	55	160	190	210	230	250	280
Lane Shift Taper Length	L_L	30	50	80	100	110	120	130	140
Downstream Taper Length	L_D	30	30	30	30	30	30	30	30
TCP, Signal, and Shoulder Taper Length (min. 5 devices)	L_S	5	8	15	15	15	15	15	15
Minimum Tangent Length between Tapers	L_T	30	60	160	190	210	230	250	280
Run-In Length on Centreline	L_R	40	50	60	60	70	80	90	100

TABLE B – DEVICE SPACING LENGTHS

Device Spacing (m)		Regular Posted Speed Limit (km/h)							
		≤50	60	70	80	90	100	110	120
Construction Sign Spacing	A	40	60	80	100	150	150	200	200
Buffer Space	B	30	40	60	80	110	140	170	200
Roll-Ahead Buffer Distance	R	30	30	40	40	40	50	50	50
Channelizing Device Spacing for Tapers	C	10	10	15	15	15	15	15	15
Channelizing Device Spacing on Curves and Tangents	D	10	10	30	30	40	40	40	50

12.1 General Information - Roundabouts[1]

1. The one-way flow and roadway geometry of roundabouts combine to make them significantly different from most other intersections negotiated by drivers.
2. Temporary traffic control during maintenance and construction activities at roundabouts should provide clear guidance to drivers, some of whom may be unfamiliar with roundabout operations.
3. Some basic features of roundabouts that are mentioned throughout this section are shown below in *Figure 12.1: Generic Two-Lane Roundabout.*

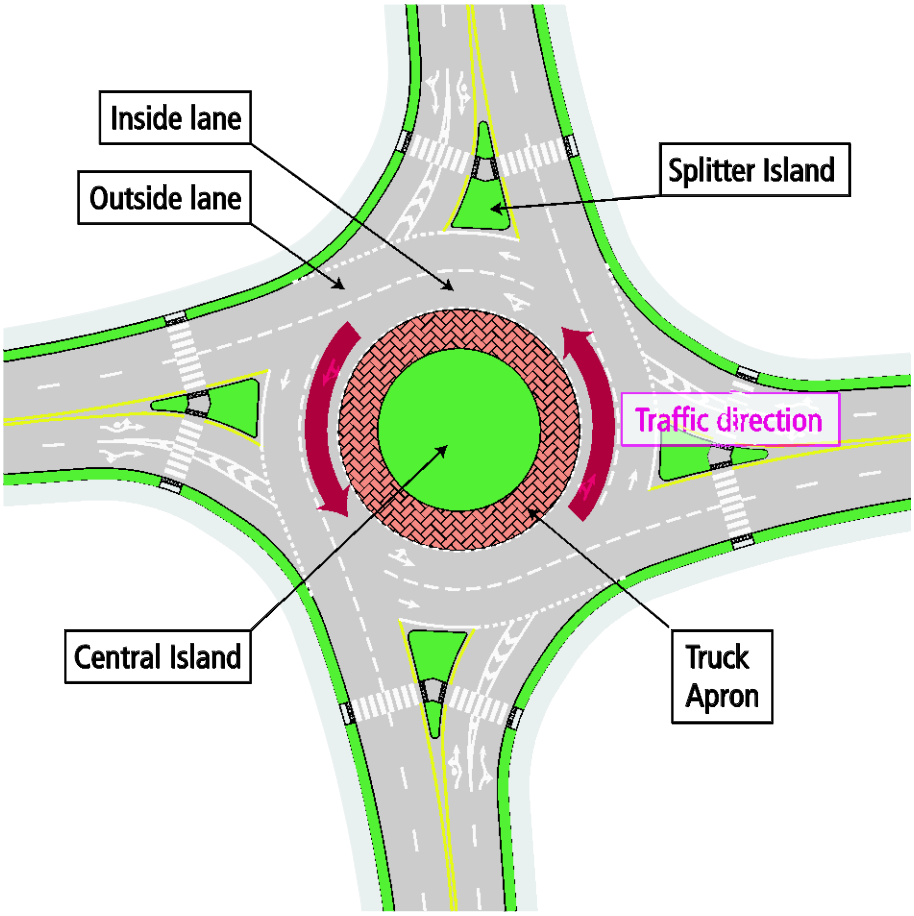


Figure 12.1: Generic Two-Lane Roundabout

[1] See *Section 11: Traffic Control Layouts – Intersections* for information on traffic control layouts for other intersection types.

The traffic control layouts are considered the minimum standard. The associated text description of each figure highlights the key standards as well as guidance and options that can be considered by the user. The following information is typical on most layouts in this section:

Standard:

- For short-duration work, a Crew Working Ahead C-004 sign shall be used.
- For long-duration work, a Construction Ahead C-018-1A sign shall be used.
- A Prepare to Stop C-029 and Traffic Control Person Ahead C-001-1 signs shall be used in advance of TCPs.
- Existing directional signs that are contrary to the new flow of traffic shall be covered.

Guidance:

- Using tubular markers rather than drums allows for better sight lines between TCPs and vehicle queues, and provides more room for large vehicles to navigate through the roundabout.

Options:

- The Flagger Ahead C-001-2 sign may be used for additional advance warning where TCPs are stopping traffic
- A portable dynamic message sign (DMS) may be used to provide advance messaging for drivers.
- In low speed (≤ 60 km/h) urban areas (within municipal boundaries), customization of traffic control layouts, including closer device spacing and shorter taper lengths, may be necessary to maintain access and mobility. In these cases, document why adjustments are being made.

12.2 Lane Closure in Roundabout – Single Lane – Short and Long Duration

Purpose:

This layout shows the typical setup where an area within the roundabout or the approach to the roundabout is obstructed, preventing traffic from entering the roundabout in its normal path. It is typically used for short-duration work, but it may also be used for long-duration work.

As is the case for traffic control at a signalized or stop-controlled intersection, a Traffic Control Person is required for each approach leg entering the roundabout because traffic is being directed in the opposite direction through the roundabout.

Standard:

- The TCPs shall have communication with one another.
- Overhead lighting shall illuminate each TCP location at night.
- The TCP on each approach leg shall hold traffic so that only one direction proceeds at a time.
- When TCPs are directing traffic, the construction speed limit shall be ≤ 70 km/h.
- Channelizing devices shall be used to isolate the work activity area and guide traffic through the roundabout.
- Barricades are required at each end of the work activity area for long-duration work.

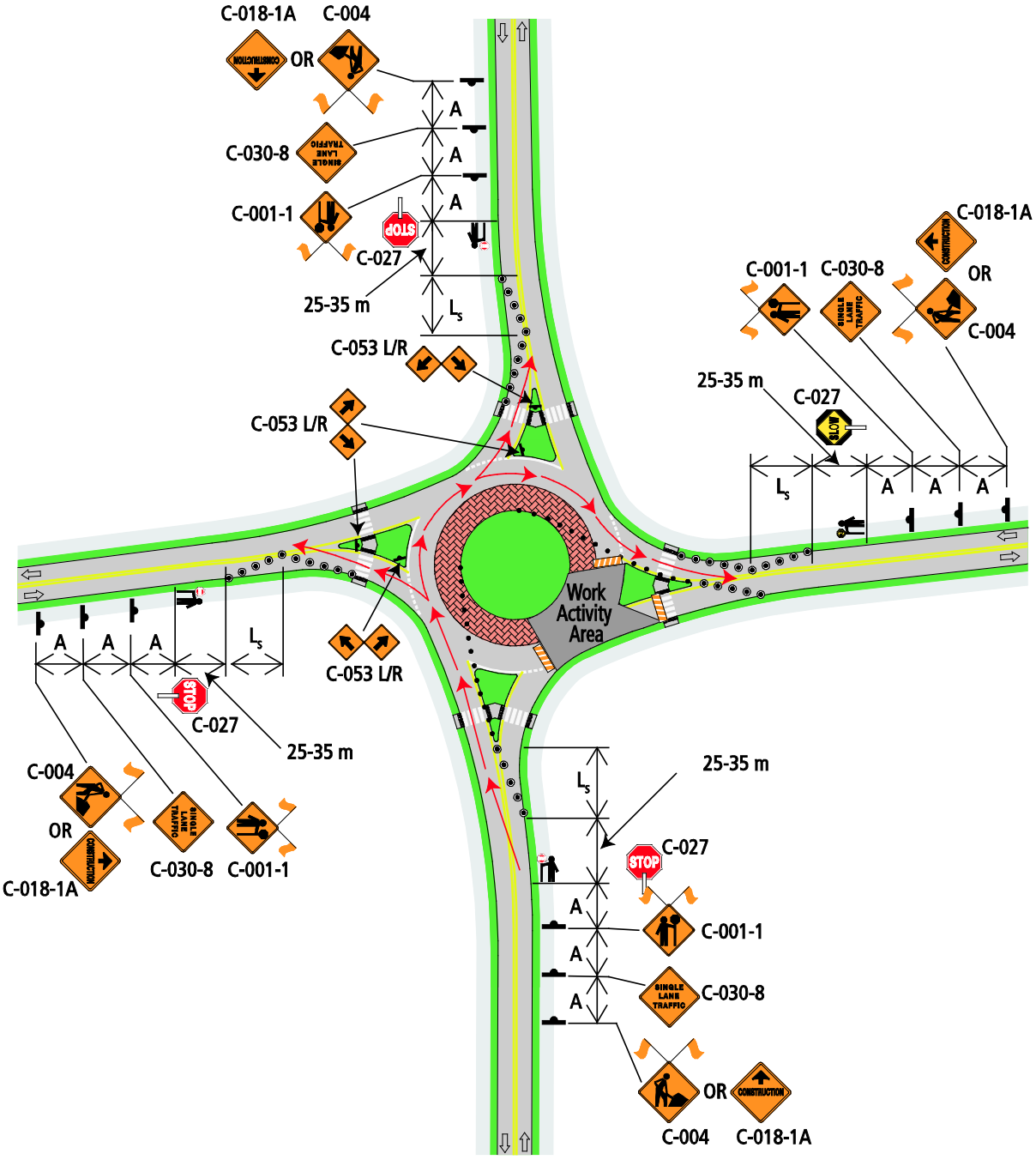
Guidance:

- Where traffic must travel counter to its normal flow because of a full closure within the roundabout, additional signing to direct drivers may be needed on splitter islands and/or within the roundabout central island (e.g., detour signs with arrows).

Options:

- Additional signing in the central island may be necessary to assist traffic movement through roundabout.
- Where approach speeds are ≤60 km/h, cones may be used instead of tubular markers.
- The truck apron may be used as part of a temporary lane to divert traffic around a lane closure within the roundabout.
- An additional TCP stationed within the central island may assist in directing drivers.
- If all the work is contained within the central island and does not affect the travel lanes, one sign per approach may be sufficient (i.e., a Crew Working Ahead C-004 sign for short-duration work or a Construction Ahead C-018-1A sign for long-duration work).

Figure 12.2: Lane Closure in Roundabout – Single Lane – Short and Long Duration



12.3 Work Outside Roundabout – Short and Long Duration

Purpose:

This layout shows the typical setup where a lane is closed on an approach to—or a departure from—a roundabout but traffic is still able to enter and proceed through the roundabout in its normal path.

As is the case for a single lane alternating setup, one Traffic Control Person is typically required on each side of the work. Advance warning signage is required on each leg entering the roundabout.

Since the closure does not affect the roundabout itself, traffic flows counter-clockwise as usual through the roundabout.

If traffic cannot use the regular entrance into the roundabout, use the layout described in *Section 12.2: Lane Closure in Roundabout – Single Lane.*

Standard:

- A Lane Closure Arrow C-053 sign shall be place on the central island to direct traffic back into the right lane.
- Channelizing devices shall be used to isolate the work area and guide traffic into the roundabout.
- Barricades are required at each end of the work activity area for long-duration work.
- When TCPs are directing traffic, the construction speed limit shall be ≤ 70 km/h.

Guidance:

- Depending on the distance between TCPs, radios may be used to improve communication.

Options:

- Where approach speeds are ≤60 km/h, cones may be used instead of tubular markers.
- In simpler situations, or where traffic volumes are low, the movement of traffic around the closure may be controlled by as few as two TCPs on the affected leg.

Figure 12.3: Work Outside Roundabout – Short and Long Duration

